Renpei Cai

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Education

University of Washington Expected Graduation: December 2027

B.S. Computer Science, GPA: 3.93/4.0 *Seattle, WA*

- Relevant Coursework: Data Structures & Parallelism, Software Design & Implementation, Hardware/Software Interface, Systems Programming, Data Management, Deep Learning, Machine Learning

Experience

University of Washington Seattle, WA

CSE 344 Teaching Assistant – Data Management *September 2026 – Present*

- Lead a weekly section, teaching **30** students SQL, relational databases, and database design.
- Debug SQL queries and guide students through database design and query-processing problems during office hours.

UW RAIVN Lab / Allen Institute for AI (Ai2) Seattle, WA

Student Researcher — Machine Learning (Computer Vision) *December 2025 – Present*

- Built distributed data pipelines for **219K** videos to fine-tune a multi-view video generation model on Kubernetes.
- Scaled PyTorch training across **8–16 GPUs** with FSDP, cross-rank sharding, and resumable checkpoints.
- Matched **state-of-the-art** baseline performance on unseen scenes with PSNR 16.31, SSIM 0.522, and LPIPS 0.465.

Seven Stars Food Seattle, WA

Software Engineering Intern *March 2026 – June 2026*

- Built a full-stack React Native/Express ordering system for a seafood wholesaler, eliminating **28 hours** of invoice entry/week with QuickBooks APIs and PostgreSQL.
- Saved **14 hours/week** in warehouse operations by automatically generating pick/pack reports from live orders.
- Powered **$180K+ in monthly orders** with transactional inventory updates, idempotent QuickBooks invoice/webhook processing, and a GitHub-to-Railway CI/CD deployment pipeline.

Algoverse Seattle, WA

Student Researcher — LLMs & Tool Usage *January 2025 – March 2025*

- **Reduced** unnecessary LLM tool calls by up to **70%** while keeping accuracy within 5% through SKT-Tool.
- Published at **NAACL Insights 2025** after building synthetic data pipelines for tool-use evaluation.
- Measured accuracy-efficiency tradeoffs across models through automated QA evaluation and ablation pipelines.

PERMIAS Seattle Seattle, WA

Volunteer Software Engineer *September 2024 – March 2025*

- Processed **5K+ transactions in 10 hours for 500+ users** through a React/Firebase event payment platform with JWT authentication, live inventory management, and real-time KPI dashboards.
- Reduced processing times by 70% by automating wallet top-up via Python email receipt parsing (IMAP API).

Projects & Leadership

Mary's Place | *Microsoft Power Pages, Dataverse, Power Automate, Power Apps, Copilot Studio, JavaScript*

- Reduced intake time by **40%** for **20+ families/day** by developing a multilingual Power Pages shelter intake system backed by Dataverse.
- Designed a normalized Dataverse schema + RBAC model for guest, household, and document data, etc.

Fitcheck | *Python, TypeScript, React, AWS Bedrock, Lambda, S3, DynamoDB, Cognito*

- Built a browser-based virtual try-on from any e-commerce page via personalized previews with AWS Bedrock.
- Supported asynchronous AI generation at scale with AWS Lambda, DynamoDB, S3, and job tracking.

Technology Consulting Association (UW) | *Director of Membership*

- Lead biannual recruitment cycles for **150+ applications** each to select new consulting cohorts.
- Coordinate 10 executive members to define evaluation criteria, review candidates, and plan recruitment events.

Technical Skills

Languages: Python, C, Java, JavaScript/TypeScript, SQL
Backend & Data: PostgreSQL, DynamoDB, Firebase, REST APIs, Node.js, Express, Flask
Frontend: React, React Native
Cloud & AI: AWS, Linux, Git, CI/CD, PyTorch, PyTorch Lightning, Distributed Training